Name

Mathematics Stage 5 Core

Negative Indices

Book 1 Apply index laws to numerical expressions with negative-integer indices

Version: 261008

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MA5-IND-C-01 simplifies algebraic expressions involving positive-integer and zero indices, and establishes the meaning of negative indices for numerical bases

Apply index laws to numerical expressions with negative-integer indices

- Apply index notation, patterns and index laws to establish the meaning of negative indices for numerical bases
- Evaluate numerical expressions involving a negative index by first representing them with a positive index
- Represent given numbers in index form (integer indices and bases only) and vice versa

The Negative Index

Review

☐ Simplify expressions using division index law

a $5^3 \div 5^7$

b $3 \div 3^5$

c $2^4 \div 2^9$

☐ Recall zero index

a 3^0

b x^0

c $3x^0$

Investigation Negative indices

Evaluate the following:

$5^3 =$

$5^2 =$

$5^1 =$

$5^0 =$

$5^{-1} =$

$5^{-2} =$

$5^{-3} =$

$3^{-1} =$

$4^{-1} =$

$5^{-1} =$

$6^{-1} =$

$3^{-2} =$

$4^{-2} =$

$5^{-2} =$

$6^{-2} =$

For any non-zero number a, $a^{-n} =$ ……

$\frac{1}{a^n}$ can be written as ……

Show why this is true using:

Division index law	Fractions
$7^3 \div 7^5 =$	$\frac{7^3 \div 7^3}{7^5 \div 7^3} =$

Index Law of Negative Powers

$$a^{-n} = \frac{1}{a^n}$$

Write negative powers as a positive power

6^{-1} $= \frac{1}{6^1}$	3^{-4} $= \frac{1}{3^4}$	10^{-3} $= \frac{1}{10^3}$

Rewrite these numbers in index form using a positive power.

a 2^{-5}	**b** 7^{-2}	**c** 4^{-3}
d 5^{-9}	**e** 3^{-100}	**f** 100^{-3}

Example Evaluate numbers with a negative power

2^{-3} $= \frac{1}{2^3}$ $= \frac{1}{8}$	10^{-2} $= \frac{1}{10^2}$ $= \frac{1}{100}$

Guided Practice

Evaluate these numbers.

a 5^{-2} $\frac{1}{25}$	**b** 10^{-3} $\frac{1}{1000}$
c 3^{-4} $\frac{1}{81}$	**d** 4^{-2} $\frac{1}{16}$

Key Ideas

1. $a^{-n} =$ .....

1 Complete these tables.

a

Expression	2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}	2^{-4}
Number	16	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8} = \frac{1}{2^3}$	$\frac{1}{16}$

÷ 2 ÷ 2 ÷ 2

b

Expression	3^4	3^3	3^2	3^1	3^0	3^{-1}	3^{-2}	3^{-3}	3^{-4}
Number	81	27	9	3	1	$\frac{1}{3}$	$\frac{1}{9}$	$\frac{1}{27}$	$\frac{1}{81}$

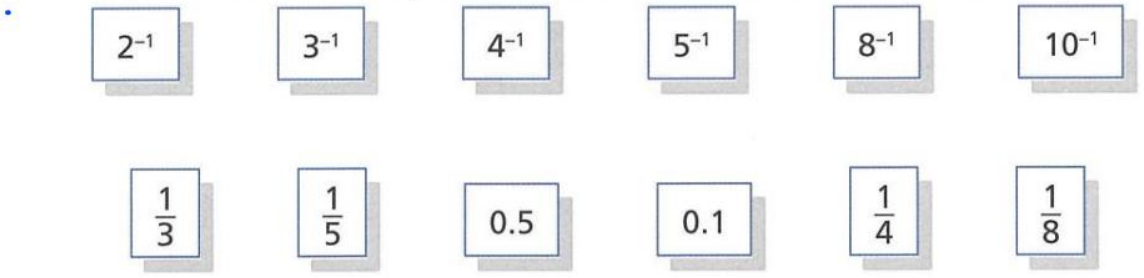
c

Expression	5^4	5^3	5^2	5^1	5^0	5^{-1}	5^{-2}	5^{-3}	5^{-4}
Number	625	125	25	5	1	$\frac{1}{5}$	$\frac{1}{25}$	$\frac{1}{125}$	$\frac{1}{625}$

d

Expression	10^4	10^3	10^2	10^1	10^0	10^{-1}	10^{-2}	10^{-3}	10^{-4}
Number	10000	1000	100	10	1	$\frac{1}{10}$	$\frac{1}{100}$	$\frac{1}{1000}$	$\frac{1}{10000}$

2 Match each card on the top row with a card from the bottom row.



2^{-1} 0.5
3^{-1} $\frac{1}{3}$
4^{-1} $\frac{1}{4}$
5^{-1} $\frac{1}{5}$
8^{-1} $\frac{1}{8}$
10^{-1} 0.1

3 Complete the table.

Power	Fraction
2^{-1}	$\frac{1}{2}$
2^{-2}	$\frac{1}{2^2} = \frac{1}{4}$
2^{-3}	
2^{-4}	
2^{-5}	

$\frac{1}{8}$
$\frac{1}{16}$
$\frac{1}{32}$

4 Complete the following:

a Simplify using index laws: $2^2 \div 2^5 = 2^{\square - \square} = 2^{\square}$

2^{-3}

b Write in expanded form: $2^2 \div 2^5 = \dfrac{2 \times 2}{\square} = \dfrac{1}{2^{\square}}$

$\frac{1}{2^3}$

c Therefore, $2^{-3} = \dfrac{1}{2^{\square}}$

$2^{-3} = \frac{1}{2^3}$

5 Complete the following:

a $10^{-4} = \dfrac{1}{10^{\square}}$

4

b $3^{-2} = \dfrac{1}{3^{\square}}$

2

c $7^{-3} = \dfrac{1}{7^{\square}}$

3

d $8^{-6} = \dfrac{1}{8^{\square}}$

6

6 Read the example carefully and answer the questions.

Madeline's work

Write as an expression without indices:

4^{-2}

4^{-2}

-16

How does a negative exponent affect its base?

Negative index means to take the reciprocal of the base

What is the correct answer to this problem?

$\frac{1}{16}$

Write as an expression without indices:

5^{-3}

$\frac{1}{125}$

7 Rewrite the following as fractions with positive indices.

a 3^{-2} $= \frac{1}{3^2}$	**b** 4^{-3} $\frac{1}{4^3}$	**c** 2^{-5} $\frac{1}{2^5}$	**d** 8^{-2} $\frac{1}{8^2}$
e 12^{-1} $\frac{1}{12}$	**f** 9^{-2} $\frac{1}{9^2}$	**g** 6^{-1} $\frac{1}{6}$	**h** 7^{-3} $\frac{1}{7^3}$
i 3^{-6} $\frac{1}{3^6}$	**j** 2^{-8} $\frac{1}{2^8}$	**k** 5^{-1} $\frac{1}{5}$	**l** 10^{-5} $\frac{1}{10^5}$

8 Complete the following to write these expressions as simplified fractions.

a $2^{-3} = \frac{1}{2^{\square}} = \frac{1}{\square}$

$\frac{1}{2^3} = \frac{1}{8}$

b $5^{-1} = \frac{1}{5^{\square}} = \frac{1}{\square}$

$\frac{1}{5^1} = \frac{1}{5}$

c $2^{-4} = \frac{1}{2^{\square}} = \frac{1}{\square}$

$\frac{1}{2^4} = \frac{1}{16}$

d $3^{-2} = \frac{1}{3^2} = \frac{1}{\square}$

$\frac{1}{9}$

e $10^{-2} = \frac{1}{10^2} = \frac{1}{\square}$

$\frac{1}{100}$

f $6^{-2} = \frac{1}{\square^{\square}} = \frac{1}{\square}$

$\frac{1}{6^2} = \frac{1}{36}$

9 Write the following as simplified fractions without powers.

a 3^{-2} $= \frac{1}{3^2} = \frac{1}{9}$

b 2^{-5} $\frac{1}{32}$

c 4^{-3} $\frac{1}{64}$

d 5^{-4} $\frac{1}{625}$

e 6^{-3} $\frac{1}{216}$

f 9^{-2} $\frac{1}{81}$

g 3^{-4} $\frac{1}{81}$

h 10^{-4} $\frac{1}{10000}$

i 2^{-6} $\frac{1}{64}$

j 5^{-3} $\frac{1}{125}$

k 6^{-2} $\frac{1}{36}$

l 10^{-3} $\frac{1}{1000}$

m $(-2)^{-1}$ $-\frac{1}{2}$

n $(-2)^{-2}$ $\frac{1}{4}$

o $-(-2)^{-2}$ $-\frac{1}{4}$

p $(-2)^{-3}$ $-\frac{1}{8}$

10 Explain why these two expressions do not have the same value.

$(-5)^{-2}$ -5^{-2}

One has the negative raised to the power and the other doesn't.

$(-5)^{-2} = \frac{1}{(-5)^2} = \frac{1}{25}$

$-5^{-2} = -\frac{1}{5^2} = -\frac{1}{25}$

11 Write the following as decimals.

a 2^{-3} 0.125

b 4^{-1} 0.25

c 10^{-2} 0.01

d 25^{-1} 0.04

e 5^{-2} 0.04

f $(-10)^{-2}$ 0.01

12 Evaluate these expressions.

a 5^{-2}

$= \frac{1}{5^2}$

$= \ldots\ldots$

b 3^{-3} $\frac{1}{27}$

c 2^{-5} $\frac{1}{32}$

d 2^{-4} $\frac{1}{16}$

e 6^{-2} $\frac{1}{36}$

f 10^{-3} $\frac{1}{1000}$

g 5^{-3} $\frac{1}{125}$

h 2^{-3} $\frac{1}{8}$

i 3^{-5} $\frac{1}{243}$

j 4^{-4} $\frac{1}{256}$

k 10^{-3} $\frac{1}{1000}$

l 10^{-5} $\frac{1}{100000}$

13 Write each pair of numbers as powers of the given base number.

a 8 and $\frac{1}{8}$ as powers of 2 $2^3, 2^{-3}$

b 16 and $\frac{1}{16}$ as powers of 4 $4^2, 4^{-2}$

c 16 and $\frac{1}{16}$ as powers of 2 $2^4, 2^{-4}$

d 25 and $\frac{1}{25}$ as powers of 5 $5^2, 5^{-2}$

e 125 and $\frac{1}{125}$ as powers of 5 $5^3, 5^{-3}$

f 625 and $\frac{1}{625}$ as powers of 5 $5^4, 5^{-4}$

g 100 and $\frac{1}{100}$ as powers of 10 $10^2, 10^{-2}$

h 1 000 000 and $\frac{1}{1\,000\,000}$ as powers of 10 $10^6, 10^{-6}$

14 Mr Ding asks his class to write down the value of 2^{-3}.
Here are some of his students' answers.



a Which of the answers are correct?

C and D

b What mistakes have the other students made?

A: a negative index doesn't mean to make a positive number negative.

B: They multiplied the power and the index

15

a If $2^{10} = 1024$, then 2^{-10} is equal to .....

$\frac{1}{1024}$

b If $3^{-5} = \frac{1}{243}$, then 3^5 is equal to .....

243

16 First change each mixed numeral or decimal to a fraction, then simplify the expression.

a $\left(1\frac{1}{2}\right)^{-1}$ $\frac{2}{3}$	**b** $\left(2\frac{2}{3}\right)^{-1}$ $\frac{3}{8}$	**c** $\left(2\frac{1}{2}\right)^{-2}$ $\frac{4}{25}$	**d** $\left(1\frac{1}{3}\right)^{-3}$ $\frac{27}{64}$
e 0.2^{-1} 5	**f** 2.25^{-1} $\frac{4}{9}$	**g** 2.5^{-2} $\frac{4}{25}$	**h** 0.05^{-2} 400

Writing Fractions with a Negative Index

Index Law of Negative Powers

$$\frac{1}{a^n} = a^{-n}$$

Rewrite fractions using a negative power

$\frac{1}{3^2} = 3^{-2}$	$\frac{1}{7} = 7^{-1}$

Rewrite these fractions using a negative power.

a $\frac{1}{5^3}$	**b** $\frac{1}{4^3}$	**c** $\frac{1}{2^6}$
d $\frac{1}{10^4}$	**e** $\frac{1}{73^{12}}$	**f** $\frac{1}{12^{73}}$

Key Ideas

1. We can write fractions using a negative power: $\frac{1}{a^n} =$ ……

1 Find the missing numbers.

a $10^{-4} = \frac{1}{10^{\square}}$ 4

b $3^{-2} = \frac{1}{3^{\square}}$ 2

c $7^{-3} = \frac{1}{7^{\square}}$ 3

d $8^{-6} = \frac{1}{8^{\square}}$ 6

e $9^{-4} = \frac{1}{9^{\square}}$ 4

f $5^{-4} = \frac{1}{5^{\square}}$ 4

2 Rewrite these fractions as expressions with negative indices.

a $\frac{1}{2}$ 2^{-1}

b $\frac{1}{2^2}$ 2^{-2}

c $\frac{1}{2^8}$ 2^{-8}

d $\frac{1}{2^5}$ 2^{-5}

e $\frac{1}{5}$ 5^{-1}

f $\frac{1}{4^3}$ 4^{-3}

g $\frac{1}{3^4}$ 3^{-4}

h $\frac{1}{5^6}$ 5^{-6}

i $\frac{1}{6}$ 6^{-1}

j $\frac{1}{7^5}$ 7^{-5}

k $\frac{1}{4^9}$ 4^{-9}

l $\frac{1}{10^2}$ 10^{-2}

3 Complete the following:

a $\frac{1}{8} = \frac{1}{2^{\square}} = 2^{\square}$ 2^{-3}

b $\frac{1}{9} = \frac{1}{3^{\square}} = 3^{\square}$ 3^{-2}

c $\frac{1}{16} = \frac{1}{2^{\square}} = 2^{\square}$ 2^{-4}

d $\frac{1}{25} = \frac{1}{5^{\square}} = 5^{\square}$ 5^{-2}

4 Write these fractions using the smallest possible index. For example, $\frac{1}{8} = \frac{1}{2^3} = 2^{-3}$

a $\frac{1}{4}$ 2^{-2}

b $\frac{1}{9}$ 3^{-2}

c $\frac{1}{125}$ 5^{-3}

d $\frac{1}{27}$ 3^{-3}

Negative Indices in the Denominator

Review

☐ Write negative indices as fractions

a	5^{-3}	**b**	3^{-4}	**c**	7^{-10}

☐ Identify the reciprocal of a number

a	$\frac{1}{5}$	**b**	3	**c**	113
d	$\frac{2}{5}$	**e**	$\frac{1}{5^3}$	**f**	6^5

Investigation Negative index in the denominator

Consider that we have a negative index in the denominator.

$$\frac{1}{5^{-3}}$$

We can find the value of this in two ways:

Find an equivalent fraction

$$\frac{1}{5^{-3}} \frac{\times 5^3}{\times 5^3}$$

Fraction division

$$\frac{1}{\left(\frac{1}{5^3}\right)} =$$

Therefore, we can conclude that:

$$\frac{1}{a^{-n}} =$$

Index Law of Negative Powers

$$\frac{1}{a^{-n}} = a^n$$

Rewrite negative power in the denominator

$\frac{1}{5^{-3}}$ $= 5^3$	$\frac{1}{3^{-4}}$ $= 3^4$	$\frac{1}{2^{-6}}$ $= 2^6$

Rewrite these expressions using positive powers

a $\frac{1}{4^{-2}}$	**b** $\frac{1}{3^{-5}}$	**c** $\frac{1}{10^{-3}}$
d $\frac{1}{7^{-4}}$	**e** $\frac{1}{35^{-67}}$	**f** $\frac{1}{67^{-35}}$

Key Ideas

1. If we have a negative index in the denominator, we can write this as: $\frac{1}{a^{-n}} =$ ……

1 Complete the following:

a $\frac{1}{3^{-4}} = 3^{\square}$ — 4

b $\frac{1}{7^{-6}} = 7^{\square}$ — 6

c $\frac{1}{8^{-3}} = 8^{\square}$ — 3

d $\frac{1}{4^{-6}} = 4^{\square}$ — 6

2 Write the following with positive indices.

a $\frac{1}{3^{-4}}$ — 3^4

b $\frac{1}{2^{-7}}$ — 2^7

c $\frac{1}{7^{-2}}$ — 7^2

d $\frac{1}{6^{-1}}$ — 6

e $\frac{1}{4^{-5}}$ — 4^5

f $\frac{1}{5^{-4}}$ — 5^4

g $\frac{1}{5^{-2}}$ — 5^2

h $\frac{1}{77^{-1}}$ — 77

3 Simplify, give your answer in positive index form.

a $\frac{1}{3^{-5}} \times 3^5$

$3^5 \times 3^5 = 3^{10}$

b $\left(\frac{1}{3^{-2}}\right)^4$

$\frac{1}{3^{-8}} = 3^8$

4 Find the value of n such that

$$\frac{1}{a^{-n}} = a^6$$

$a^n = a^6$

$n = 6$

5 Find the value of n such that

$$\frac{1}{a^{-n}} = a^{-6}$$

$a^n = a^{-6}$

$n = -6$

6 Simplify, writing with positive indices

$$\frac{1}{7^{-3} \times 7^{-2}}$$

$\frac{1}{7^{-5}} = 7^5$

Mixed Practice

ⓘ Index Laws of Negative Indices

$a^{-n} = \frac{1}{a^n}$	$\frac{1}{a^{-n}} = a^n$

1 Rewrite with a positive index. Do not evaluate.

a 2^{-1}

$\frac{1}{2}$

b 2^{-3}

$\frac{1}{2^3}$

c 3^{-2}

$\frac{1}{3^2}$

d 5^{-2}

$\frac{1}{5^2}$

e 5^{-10}

$\frac{1}{5^{10}}$

f 10^{-5}

$\frac{1}{10^5}$

2 Rewrite with a negative power. Do not evaluate.

a $\frac{1}{10^2}$

10^{-2}

b $\frac{1}{10^3}$

10^{-3}

c $\frac{1}{5^3}$

5^{-3}

d $\frac{1}{2^3}$

2^{-3}

e $\frac{1}{3^2}$

3^{-2}

f $\frac{1}{100^2}$

100^{-2}

3 Rewrite with a positive power. Do not evaluate.

a $\frac{1}{4^{-3}}$

4^3

b $\frac{1}{4^{-1}}$

4

c $\frac{1}{4^{-10}}$

4^{10}

d $\frac{1}{7^{-10}}$

7^{10}

e $\frac{1}{10^{-7}}$

10^7

f $\frac{1}{10^{-1}}$

10

4 Write with a positive power. Do not evaluate.

a 2^{-1} $\frac{1}{2}$	**b** $\frac{1}{2^{-1}}$ 2^{-1}	**c** 2^{-3} $\frac{1}{2^{-3}}$	**d** $\frac{1}{2^{-3}}$ 2^{3}
e 5^{-3} $\frac{1}{5^{3}}$	**f** $\frac{1}{5^{-3}}$ 5^{3}	**g** $\frac{1}{7^{-3}}$ 7^{3}	**h** 7^{-3} $\frac{1}{7^{3}}$
i 7^{-5} 7^{5}	**j** 12^{-5} $\frac{1}{12^{5}}$	**k** 5^{-12} $\frac{1}{5^{12}}$	**l** $\frac{1}{5^{-12}}$ 5^{12}
m $\frac{1}{12^{-5}}$ 12^{5}	**n** $\frac{1}{7^{-6}}$ 7^{6}	**o** 6^{-7} $\frac{1}{6^{7}}$	**p** $\frac{1}{6^{-7}}$ 6^{7}

5 Evaluate, without using a calculator.

a 3^{-2} $\frac{1}{9}$	**b** $\frac{1}{5^{3}}$ $\frac{1}{125}$	**c** $\frac{1}{4^{2}}$ $\frac{1}{16}$	**d** 2^{-4} $\frac{1}{16}$
e $\frac{1}{3^{-1}}$ 3	**f** $\frac{1}{2^{3}}$ $\frac{1}{8}$	**g** 5^{-1} $\frac{1}{5}$	**h** $\frac{1}{10^{-2}}$ 100
i $\frac{1}{6^{2}}$ $\frac{1}{36}$	**j** 4^{-2} $\frac{1}{16}$	**k** $\frac{1}{2^{-4}}$ 16	**l** $\frac{1}{5^{3}}$ $\frac{1}{125}$
m 10^{-1} $\frac{1}{10}$	**n** $\frac{1}{3^{-3}}$ 27	**o** $\frac{1}{4^{1}}$ $\frac{1}{4}$	**p** 6^{-2} $\frac{1}{36}$
q $\frac{1}{10^{-3}}$ 1000	**r** $\frac{1}{3^{2}}$ $\frac{1}{9}$	**s** 2^{-3} $\frac{1}{8}$	**t** $\frac{1}{6^{-1}}$ 6

Negative Indices with Coefficients

ⓘ Index Laws of Negative Indices

When a coefficient multiplies an expression with a negative index, apply the rule to the number with a negative index only.

$$c \times a^{-n} = \frac{c}{a^n}$$

$$\frac{c}{a^{-n}} = c \times a^n$$

ⓘ **Rewrite** using a positive index

4×3^{-2}

$= 4 \times \frac{1}{3^2}$

$= \frac{4}{3^2}$

4×3^{-5}

$= 4 \times \frac{1}{3^5}$

$= \frac{4}{3^5}$

10×8^{-5}

$= 10 \times \frac{1}{8^5}$

$= \frac{10}{8^5}$

Write using a positive index

a 3×5^{-9}

b 2×3^{-9}

c -4×2^{-5}

d $5^{-3} \times 2$

e $7^{-2} \times 3$

f -3×2^{-1}

ⓘ **Rewrite** using a negative index

$\frac{5}{3^2}$

$= 5 \times 3^{-2}$

$\frac{5}{3^9}$

$= 5 \times 3^{-9}$

$\frac{12}{5^9}$

$= 12 \times 5^{-9}$

Write using a positive index

a $\frac{3}{4^2}$

b $\frac{9}{4^2}$

c $\frac{-1}{4^2}$

d $-\frac{1}{4^2}$

e $\frac{-7}{13^5}$

f $\frac{7}{13^5}$

Rewrite using a positive index

$\frac{5}{3^{-2}}$	$\frac{5}{6^{-2}}$	$\frac{11}{6^{-7}}$
$= 5 \times 3^2$	$= 5 \times 6^2$	$= 11 \times 6^7$

Write using a positive index

a $\frac{3}{4^{-2}}$	**b** $\frac{6}{5^{-3}}$	**c** $\frac{4}{3^{-5}}$
d $\frac{-4}{3^{-5}}$	**e** $\frac{-7}{13^{-5}}$	**f** $\frac{12}{5^{-1}}$

Example Evaluate expressions with negative indices

$\frac{3}{2^{-1}}$

$= 3 \times 2$

$= 6$

4×3^{-2}

$= \frac{4}{3^2}$

$= \frac{4}{9}$

Guided Practice

a 3×5^{-2}

$\frac{3}{25}$

b $\frac{5}{3^{-2}}$

45

c $\frac{3}{4^{-2}}$

48

d 7×3^{-2}

$\frac{7}{9}$

Key Ideas

1. $5 \times 3^{-7} =$
2. $\frac{5}{3^7} =$
3. $\frac{5}{3^{-7}} =$

1 Write the following as simplified fractions. Leave your answer with indices.

a 3×2^{-3}

$= 3 \times \frac{1}{2^3}$

$= \ldots\ldots\ldots\ldots$

$\frac{3}{2^3}$

b 5×2^{-3}

$\frac{5}{2^3}$

c -2×5^{-2}

$-\frac{2}{5^2}$

d 5×3^{-4}

$\frac{5}{3^4}$

e 12×3^{-5}

$\frac{12}{3^5}$

f 9×7^{-2}

$\frac{9}{7^2}$

g 2×7^{-5}

$\frac{2}{7^5}$

h -3×5^{-4}

$-\frac{3}{5^4}$

2 Rewrite the following with positive indices.

a $\frac{6}{4^{-3}}$

6×4^3

b $\frac{-5}{8^{-2}}$

-5×8^2

c $\frac{4}{7^{-5}}$

4×7^5

d $\frac{3}{2^{-5}}$

3×2^5

e $\frac{12}{5^{-4}}$

12×5^4

f $\frac{6}{10^{-3}}$

6×10^3

g $\frac{-7}{9^{-4}}$

-7×9^4

h $\frac{7}{10^{-6}}$

7×10^6

3 Write the following as simplified fractions with no indices.

a 3×2^{-3}

$= 3 \times \frac{1}{8}$

$= \ldots\ldots\ldots\ldots$

b 5×4^{-3}

$\frac{5}{64}$

c 7×5^{-2}

$\frac{7}{25}$

d -2×3^{-4}

$-\frac{2}{81}$

e 4×3^{-5}

$\frac{4}{243}$

f -9×5^{-2}

$-\frac{9}{25}$

g 8×7^{-3}

$\frac{8}{243}$

h 6×5^{-2}

$\frac{6}{25}$

4 Write as a simplified fraction.

a 5×10^{-4}

$\frac{5}{10000} = \frac{1}{2000}$

b 1.5×10^{-3}

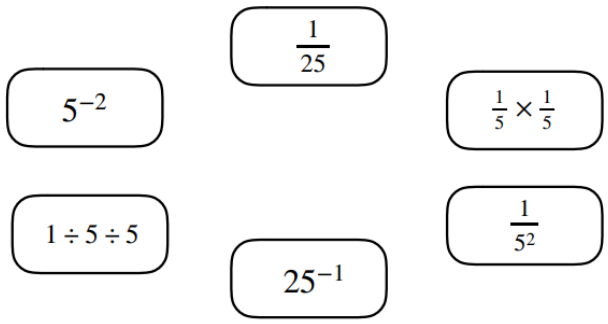
$\frac{1.5}{1000} = \frac{3}{2000}$

c 3.42×10^{-6}

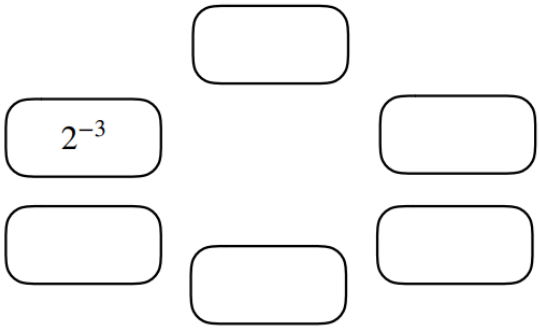
$\frac{3.42}{1\,000\,000} = \frac{1.71}{500\,000} = \frac{171}{50\,000\,000}$

7 Complete each collection of equivalent expressions.

a *Example*



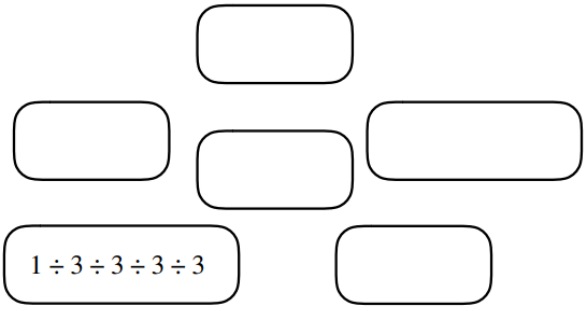
b



Clockwise from top:

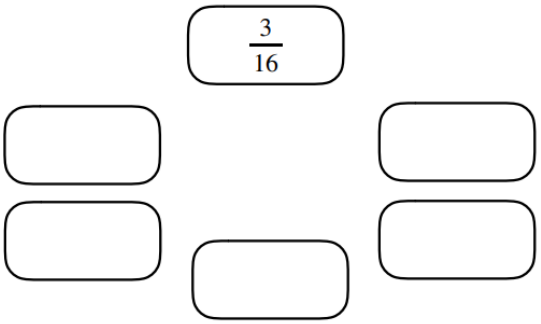
$\frac{1}{8}$, $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$, $\frac{1}{2^3}$, 8^{-1}, $1 \div 2 \div 2 \div 2$

c



$\frac{1}{81}$, $\frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3}$, $\frac{1}{3^4}$, 81^{-1}, 3^{-4}

d



$3 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$, $\frac{3}{2^4}$, 3×2^{-4}, $3 \div 2 \div 2 \div 2 \div 2$, 3×2^{-4}

8 **NAPLAN A**

Select the expression that is equivalent to $(4^{-3} \div 4^6)^3$

A. 4^{-27}

B. 4^{-9}

C. 4^{-54}

D. 4^{-6}

$(4^{-9})^3$

$= 4^{-27}$

A

9 Evaluate the following without using a calculator. Write your answer without indices.

a 5^{-1} $\frac{1}{5}$	**b** 3^{-2} $\frac{1}{9}$	**c** $(-4)^{-2}$ $\frac{1}{16}$
d 4×10^{-2} $\frac{1}{25}$	**e** -5^{-2} $-\frac{1}{25}$	**f** -5×10^{-3} $-\frac{1}{200}$
g $6^{4} \times 6^{-6}$ $\frac{1}{36}$	**h** $8^{-7} \times (8^{2})^{3}$ $\frac{1}{8}$	**i** $(5^{2})^{-1} \times (2^{-2})^{-1}$ $\frac{4}{25}$
j $\frac{1}{8^{-1}}$ 8	**k** $\frac{1}{10^{-2}}$ 100	**l** $\frac{-2}{5^{-3}}$ -250
m $\frac{2}{2^{-3}}$ 16	**n** $\frac{-5}{2^{-1}}$ -10	**o** $\frac{2^{3}}{2^{-3}}$ 64
p $\left(\frac{3}{8}\right)^{-2}$ $\frac{64}{9}$	**q** $\left(\frac{-5}{2}\right)^{2}$ $\frac{25}{4}$	**r** $\frac{(3^{-2})^{3}}{3^{-5}}$ $\frac{1}{3}$